Maharaja Ranjit Singh Punjab Technical University
Bathinda-151001



FACULTY OF ENGINEERING & TECHNOLOGY

SYLLABUS

FOR

B.TECH. COMPUTER SCIENCE & ENGINEERING

(4 YEARS PROGRAMME)

2022 BATCH ONWARDS

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(ii) Subject to change in the syllabi at any time.

Please visit the University website time to time.

(7th SEMESTER)

Course		Contact Hrs.			Marks			Credits
Code	**Name**	**L**	**T**	**P**	**Int.**	**Ext.**	**Total**	
BCSES1-701	*Project-II	0	0	4	60	40	100	2
BCSES1-702	***Industrial Training-III	-	-	-	60	40	100	4
	Departmental Elective-IV	3	0	0	40	60	100	3
BCSED1-711	Distributed Operating System							
BCSED1712	Soft Computing							
BCSED1-713	Human Computer Interaction							
BCSED1-714	Ad-hoc & Sensor Networks							
	Departmental Elective-V	3	0	0	40	60	100	3
BCSED1-721	Bioinformatics							
BCSED1-722	Image processing							
BCSED1-723	Cryptography & Network Security							
BCSED1-724	Artificial Intelligence							
XXXX	**Open Elective***	3	0	0	40	60	100	3
BMNCC0-002	Environmental Sciences	2	0	0	100	00	100	0
	Mandatory Courses- noncredit****	2	0	0	100	00	100	0
BMNCC0-001	Constitution of India							
BMNCC0-006	Essence of Indian Knowledge Tradition							
Total		**-**	**-**	**-**	**440**	**260**	**700**	**15**

*Open Elective Subjects may also be chosen from the list of Open Electives-I, II and III offered by other departments of university.

****Continued from VII Semester, Project work, seminar and internship in industry or at appropriate work place

*****During the summer vacation after 6th semester.

***choose any one subject from mandatory Courses.

(8th SEMESTER)

Course		Contact Hrs.			Marks			Credits
Code	**Name**	**L**	**T**	**P**	**Int.**	**Ext.**	**Total**	
BCSES1-801	Project-III**	0	0	10	60	40	100	5
	Departmental Elective-VI	3	0	0	40	60	100	3
BCSED1-811	Enterprise Resource Planning							
BCSED1-812	Internet of things							
BCSED1-813	Advanced Database Management Systems							
BCSED1-814	Software Project Management							
XXXX	**Open Elective***	3	0	0	40	60	100	3
XXXX	**Open Elective***	3	0	0	40	60	100	3
Total		**-**	**-**	**-**	**180**	**220**	**400**	**14**

*Open Elective Subjects may also be chosen from the list of Open Electives-I, II and III offered by other departments of university.

**Project III to be made by student during the semester.

SOFT COMPUTING

Subject Code- BCSED1-712 **L T P C** **Duration – 45Hrs**
3 0 0 3

COURSE OBJECTIVE:
1. To introduce soft computing concepts and techniques and foster their abilities in designing appropriate technique for a given scenario.
2. To implement soft computing based solutions for real-world problems.

COURSE OUTCOMES:
1. Identify and describe soft computing techniques and their roles in building intelligent machines
2. To have knowledge of neural networks-I
3.To have knowledge of neural networks-II.
4. To learn the concepts of genetic algorithms.

COURSE CONTENTS:

UNIT-I (12 Hrs)

Introduction to Soft Computing and Neural Networks: Introduction to soft computing, soft computing constituents, difference between soft computing and hard computing, Applications of Soft Computing.
Fuzzy Logic: Basic Concepts, Fuzzy Sets and Operations, Properties of Fuzzy Sets, Fuzzy Relations, Membership Functions, Fuzzy Rules and Fuzzy Reasoning, Fuzzy Inference Systems, Defuzzification methods, Industrial applications.

UNIT-II (10 Hrs)

Neural Networks-I: (Introduction & Architecture): Biological Neuron, Machine Learning Using Neural Network, Artificial Neuron and its model, activation functions, Supervised, unsupervised and reinforcement Learning, feed forward networks and feedback networks, learning rules – Hebbian, Delta, Perceptron learning and Windrow-Hoff, winner-take-all.

UNIT-III (12 Hrs.)

Neural Networks-II: Supervised learning- Perceptron learning, single layer/multilayer perceptron, linear separability, hidden layers, back propagation algorithm, Radial Basis Function network; Unsupervised learning - Kohonen, SOM, k-means clustering, Adaptive Resonance Theory (ART), Application of neural networks.

UNIT-IV (11 Hrs)

Genetic Algorithms: Concept of Introduction to Genetic Algorithms (GA),GA operators: Encoding, Crossover, Selection, Mutation, Fitness function, population, Simple GA (SGA), other types of GA,Applications of GA .

Recommended Books:
1. Jyh: Shing Roger Jang, Chuen:Tsai Sun, EijiMizutani, „Neuro: Fuzzy and Soft Computing17‟, Prentice-Hall of India, 2003.
2. George J. Klir and Bo Yuan, „Fuzzy Sets and Fuzzy Logic: Theory and Applications17‟, Prentice Hall, 1995. 3. MATLAB Toolkit Manual

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HUMAN COMPUTER INTERACTION

Subject Code- BCSED1-713 **L T P C** **Duration – 45 Hrs.**
3 0 0 3

COURSE OBJECTIVE:

1. Describe and apply core theories, models and methodologies from the field of HCI
2. Discuss current research in the field of HCI

COURSE OUTCOMES:

1. To have knowledge of task centred systems design.
2. Understand the fundamental aspects of designing and evaluating interfaces
3. To understand different design principles.
4. To learn different HCI design standards.

COURSE CONTENTS:

UNIT-I (11 Hrs)

Introduction, Task-centred system design, User-centred design and prototyping: Human-Computer Interaction. Task-centred system design: Task-centered process, development of task examples, evaluation of designs through a task-centered walk-through. **User-centred design and prototyping**: Assumptions, participatory design, methods for involving the user, prototyping, low fidelity prototypes, medium fidelity

UNIT-II (12 Hrs)

Methods for evaluation of interfaces with users and Psychology of everyday things: Goals of evaluation, approaches, ethics, introspection, extracting the conceptual model, direct observation, constructive interaction, interviews and questionnaires, continuous evaluation via user feedback and field studies, choosing an evaluation method.

Psychology of everyday things: Psychopathology of everyday things, examples, concepts for designing everyday things. Beyond screen design: characteristics of good representations, information visualization, Tufte‟s guidelines, visual variables, metaphors, direct manipulation

UNIT-III (11 Hrs)

Graphical screen design, Design principles and usability heuristics: Graphical design concepts, components of visible language, graphical design by grids. Design principles and usability heuristics: Design principles, principles to support usability, golden rules and heuristics, HCI patterns

UNIT-IV (11 Hrs)

HCI design standards, Past and future of HCI: Process-oriented standards, product-oriented standards, strengths and limitations of HCI Standards. Past and future of HCI: The past, present and future, perceptual interfaces, context-awareness and perception

Recommended Books:

1. Dix A., Finlay J., Abowd G. D. and Beale R., Human Computer Interaction, Pearson Education, 3rd edition, 2005.
2. Preece J., Rogers Y., Sharp H., Baniyon D., Holland S. and Carey T. Human Computer Interaction, Addison Wesley, 1st edition, 1994.

CRYPTOGRAPHY & NETWORK SECURITY

Subject Code- BCSED1-723 **L T P C** **Duration – 45Hrs**
3 0 0 3

COURSE OBJECTIVE:
The main objective of this course is to make student able to understand the basic concepts, services, threats and principles in network security, various security services and mechanisms in the network protocol stack.

COURSE OUTCOMES:
1. To understand security trends.
2. To implement various cryptographic algorithms.
3. To implement public key cryptography.
4. To implement IP Security.

COURSE CONTENTS:

UNIT-I (12 Hrs)
Security trends, Attacks and services, Classical crypto systems, Different types of ciphers, LFSR sequences, Basic Number theory, Congruence, Chinese Remainder theorem, Modular exponentiation, Fermat and Euler's theorem, Legendre and Jacobi symbols, Finite fields, continued fractions.

UNIT-II (09 Hrs)
Simple DES, Differential crypto analysis, DES – Modes of operation, Triple DES, AES, RC4, RSA, Attacks – Primality test – factoring.

UNIT-III (12 Hrs)
Discrete Logarithms, Computing discrete logs, Diffie-Hellman key exchange, ElGamal Public key cryptosystems, Hash functions, Secure Hash, Birthday attacks, MD5, Digital signatures, RSA, ElGamal DSA.

UNIT-IV (12 Hrs)
Authentication applications – Kerberos, X.509, PKI – Electronic Mail security – PGP, S/MIME – IP security – Web Security – SSL, TLS, SET. Intruders, Malicious software, viruses and related threats, Firewalls, Security Standards.

RECOMMENDED BOOKS:
1. Wade Trappe, Lawrence C Washington, „Introduction to Cryptography with Coding Theory", 2nd Edn., Pearson, **2007**.
2. William Stallings, „Cryptography and Network Security Principles and Practices", 4th Edn., Pearson/PHI, **2006**.
3. W. Mao, „Modern Cryptography – Theory and Practice", 2nd Edn., Pearson Education, **2007**.
4. Charles P. Pfleeger, Shari Lawrence Pfleeger, „Security in Computing", 3rd Edn., Prentice Hall of India, **2006**.
5. Behrouz Forouzan, „Cryptography & Network Security", 2nd Edn., McGraw-Hill, **2011.**

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ARTIFICIAL INTELLIGENCE

Subject Code- BCSED1-724 **L T P C** **Duration – 45Hrs**
3 0 0 3

COURSE OBJECTIVE:

1. Introduce the basic principles in artificial intelligence research.
2. Explore Areas of application such as knowledge representation, natural language processing, expert systems, vision and robotics.

COURSE OUTCOMES:

1. Understand the concept of Artificial intelligence, problem solving and various types of search strategies.
2. Understand the concept of Knowledge base, knowledge representation, AI languages & tools and various planning techniques.
3. Identify uncertainty and understand fuzzy logic concept to handle uncertainty.
4. Understand the COURSE of AI agents and various COURSE methods it also includes neural network and includes the communication of AI agents and natural language processing.

COURSE CONTENTS:

UNIT-I (12 Hrs)

Introduction: History of AI - Intelligent agents – AI and Applications - Problem spaces and search - Heuristic Search techniques – Best-first search – Informal search strategies-A* algorithm, Iterative deepening A*(IDA), small memory A*(SMA). Game Playing: Minimax search procedure - Adding alpha-beta cutoffs

UNIT-II (12 Hrs)

Knowledge Representation: Approaches and issues in knowledge representation Knowledge - Based Agent- Propositional Logic – Predicate logic –Reasoning, AI languages Prolog, Lisp.

UNIT-III (09 Hrs)

Reasoning under uncertainty: Implementation- Basic probability notation - Bayes rule – Certainty factors and rule based systems - Bayesian networks, Fuzzy Logic.

UNIT IV (12 Hrs)

Planning and COURSE: Basic representation of plans - conditional planning - Multi-Agent planning. Forms of COURSE - inductive COURSE - Reinforcement COURSE - COURSE decision trees - Neural Networks. Communication: Natural language processing, Formal Grammar, Parsing

RECOMMENDED BOOKS:

1. Elaine Rich, Kevin Knight and Shivashankar B.Nair, „Artificial Intelligence“, 3rd Edn., Tata McGraw-Hill, **2009**.

2. Stuart J. Russell and Peter Norvig, „Artificial Intelligence: A Modern Approach“, Pearson Education Asia, 2nd Edn., **2003**.

3. N.P. Padhy, „Artificial Intelligence and Intelligent System“, Oxford University Press, 2nd Edn., **2005**.

4. Rajendra Akerkar, „Introduction to Artificial Intelligence“, Prentice-Hall of India, **2005**.

5. Patrick Henry Winston, „Artificial Intelligence“, Pearson Education Inc., 3rd Edn., **2001**.

6. Eugene Charniak and Drew Mc Dermott, „Introduction to Artificial Intelligence", Addison-Wesley, ISE Reprint, **1998**.
7. Nils J. Nilsson, „Artificial Intelligence - A New Synthesis", Harcourt Asia Pvt. Ltd., Morgan Kaufmann, **1988**.

ENVIRONMENTAL SCIENCES

Subject Code: BMNCC0-002 **L T P C** **Duration: 30 Hrs.**

2 0 0 0

COURSE OBJECTIVES:

1. To identify global environmental problems arising due to various engineering/industrial and technological activities and the science behind these problems.
2. To identify the major pollutants and abatement devices for environmental management and sustainable development.
3. To estimate the current world population scenario and calculating the economic growth, energy requirement, demand and also their related problems and plausible solutions.

COURSE OUTCOMES:

1. Students are able to identify global environmental problems arising due to various engineering/industrial and technological activities and the science behind these problems.
2. Students are able to classify the major pollutants and abatement devices for environmental management and sustainable development.
3. Students can evaluate the current world population scenario and calculating the economic growth, energy requirement, demand and also their related problems and plausible solutions.

. UNIT-I

1. The Multidisciplinary Nature of Environmental Studies:

Definition, scope and importance, Need for public awareness.

2. Natural Resources

Renewable and Non-renewable Resources: Natural resources and associated problems.

(a) Forest resources: Use and over-exploitation, deforestation, case studies. Timber extraction, mining, dams and their effects on forests and tribal people.

(b) Water resources: Use and over-utilization of surface and ground water, floods, drought, conflicts over water, dams-benefits and problems.

(c) Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies.

(d) Food resources: World food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity, case studies.

(e) Energy resources: Growing energy needs, renewable and non-renewable energy sources, use of alternate energy sources, case studies.

UNIT-II

Environmental Pollution: Definition

(a) Causes, effects and control measures of:

i) Air pollution
ii) Water pollution
iii) Soil pollution
iv) Marine pollution
v) Noise pollution
vi) Thermal pollution
vii) Nuclear pollution

(b) **Solid Waste Management**: Causes, effects and control measures of urban and industrial wastes.

(c) Role of an individual in prevention of pollution.

(d) Pollution Case Studies.

(e) Disaster management: floods, earthquake, cyclone and landslides.

UNIT-III

Social Issues and the Environment

(a) From unsustainable to sustainable development

(b) Urban problems and related to energy

(c) Water conservation, rain water harvesting, Watershed Management

(d) Resettlement and rehabilitation of people; its problems and concerns, Case studies.

(e) Environmental ethics: Issues and possible solutions

(f) Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust, Case studies.

(g) Issues involved in enforcement of environmental legislation

UNIT-IV

Human Population and the Environment

(a) Population growth, variation among nations

(b) Population explosion – Family Welfare Programmes

(c) Environment and human health

(d) Human Rights

(e) Value Education

(f) Women and Child Welfare

(g) Role of Information Technology in Environment and Human Health

(h) Case Studies.

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Environmental Science related activities:

We as human being are not an entity separate from the environment around us rather we are a constituent seamlessly integrated and co-exist with the environment around US. We are not an entity so separate from the environment that we can think of mastering and controlling it rather we must understand that each and every action of ours reflects on the environment and vice versa. Ancient wisdom drawn from Vedas about environment and its sustenance reflects these ethoses. There is a direct application of this wisdom even in modern times. Idea of an activity based course on environment protection is to sensitize the students on the above issues through following two types of activities.

(a) Awareness Activities:

i) Small group meetings about water management, promotion of recycle use, generation of less waste, avoiding electricity waste.

ii) Slogan making event

iii) Poster making event

iv) Cycle rally

v) Lectures from experts.

(b) Actual Activities:

i) Plantation

ii) Gifting a tree to see its full growth

iii) Cleanliness drive

iv) Drive for segregation of waste

v) To live some big environmentalist for a week or so to understand his work

vi) To work in kitchen garden for mess

vii) To know about the different varieties of plants

viii) Shutting down the fans and ACs of the campus for an hour or so

Recommended Books

1. Agarwal, K. C. 2001 Environment Biology, Nidi Publ. Ltd. Bikaner.
2. Jadhav, H & Bhosale, V.M. 1995. Environment Protection and Laws. Himalaya Pub House, Delhi 284p.
3. Rao M. N. & Datta A.K. 1987. Waste Water Treatment. Oxford & IBH Publ. Co. Pvt. Ltd. 345 p.
4. Principle of Environment Science by Cunninghan, W.P.
5. Essentials of Environment Science by Joseph.

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CONSTITUTION OF INDIA

Subject Code- BMNCC0-001 **L T P C**
2 0 0 0

COURSE OBJECTIVE:
The student will be able to learn different perspectives of constitution of India.

COURSE OUTCOMES:
1. To learn the meaning and historical perspective of law.
2. To have deep knowledge of fundamental rights.
3. To learn different policies implemented by Constitution of India.
4. To learn Article 19 and 21.

COURSE CONTENTS:
1. Meaning of the constitution law constitutionalism.
2. Historical perspective of the Constitution of India.
3. Salient features and characteristics of the Constitution of India.
4. Scheme of the fundamental rights.
5. The scheme of fundamental duties and its legal status.
6. The Directive principles of State Policy – Its importance and implementation.
7. Federal structure and distribution of legislative and financial powers between the Union and states.
8. Parliamentary form of Government of India- The Constitution powers and status of the President of India.
9. Amendment of Constitutional Powers and Procedure.
10. The historical perspectives of constitutional amendments in India.
11. Emergency Provisions: National Emergency, President Rule, financial emergency.
12. Local Self Government- Constitutional Scheme in India.
13. Scheme of Fundamental Right to Equality.
14. Scheme of the Fundamental Right to certain Freedom under Article 19
15. Scope of the Right to Life and Personal Liberty under Article 21

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UNIT-I (12 Hours)

GENERAL MANAGEMENT: Introduction and characteristics of management, Principle and function management, Scientific of management.

CONSTRUCTION MANAGEMENT: Definition, functions and scope of construction management; scientific methods of management, construction team.

UNIT-II (11 Hours)

MANPOWER PLANNING: Manpower Planning process, Role of HR manager, Personnel Principles, Managerial Staffing, Recruitment Selection strategies.

HUMAN RELATIONS AND ORGANISATIONAL BEHAVIOUR: Introduction & Significance, Basic individual psychology, Conflicts in organizations, Engineer as Manager, Communication and negotiation skills.

UNIT-III (11 Hours)

MATERIALS MANAGEMENT: Scope, Objective and functions of material management, Procurement and store management, Materials handling management, Inventory control and management, Disposal of Surplus Materials.

CONSTRUCTION PLANNING: Basic Concepts in the Development of Construction Plans, Choice of Technology and Construction Method, Estimating Activity Durations & Resource Requirements for Work Activities.

UNIT-IV (11 Hours)

CONSTRUCTION CONTRACTS: Indian Contracts Act, Elements of Contracts, Types of Contracts, Features, Suitability, Design of Contract Documents, International Contract Document, Standard Contract Document.

SITE LAYOUT: Principles governing site lay out, factors effecting site lay out, preparation of site lay out, Feasibility study, project reports, progress reports, monitoring and controlling construction activities.

CELLULAR & MOBILE COMMUNICATION

Subject Code: BECEO1-002 | **L T P C** 3 0 0 3 | **Duration: 45 Hrs.**

Course Objectives:

The student should be made to:

1. Know the characteristic of cellular mobile systems
2. Learn the various elements of cellular radio systems design and interference.
3. Understand the difference between different wireless generations.
4. Understand the various handoff techniques.

Course Outcomes:

At the end of the Course, the student should be able to

1. Understand cellular wireless communication systems.
2. Learn about elements of cellular radio systems.
3. Compare various cellular mobile generations and analyze their performance
4. Describe about hand offs and call drops.

UNIT-I (12 Hrs.)

Introduction to Cellular Mobile Systems Its Generations: A basic cellular system, performance criteria, Uniqueness of mobile radio environment, operation of cellular systems, planning a cellular system, analog & digital cellular systems. Introduction to different generations of cellular mobile systems:- first generation, second generation, third generation, fourth generation and fifth generation cellular systems.

UNIT-II (11 Hrs.)

Hand Off & Dropped Calls: Introduction to concept of handoff, need of handoff, types of handoff and their characteristics, dropped call rates & their evaluation. Introduction to Operational Techniques: Parameters, coverage hole filler, leaky feeders, cell splitting and small cells, narrow beam concept.

UNIT-III (10 Hrs.)

Elements of Cellular Radio Systems Design: General description of the problem, concept of frequency reuse channels, co-channel interference, cell splitting, consideration of the components of cellular systems.

UNIT-IV (12 Hrs.)

Cell Coverage for Signal & Traffic: General introduction, obtaining the mobile point to point mode, Introduction to the concepts of :- propagation over water or flat open area, foliage loss, propagation near in distance, long distance propagation, point to point prediction model, cell site, antenna heights and signal coverage cells, mobile to mobile propagation.

Recommended Text Books / Reference Books:

1. C.Y. Lee William,'Mobile Cellular Telecommunications', McGraw Hill.
2. KamiloFeher, 'Wireless and Digital Communications', PHI.
3. T.S. Rappaport, 'Wireless Communication, Principles & Practice', PHI.